Artificial Intelligence (CSC-203)

Objectives:

- **Discussing course contents**
- **Course Learning Objectives**
- **Definition of Artificial Intelligence**
- **Views of AI**
- **Measurements of AI**

Computer Science Core Courses

Compiler Construction	3 (3-0)
Comp. Organization & Assembly Language	4 (3-2)
Digital Logic Design	4 (3-2)
Design & Analysis of Algorithms	3 (3-0)
Parallel & Distributed Computing	3 (3-0)
Artificial Intelligence	4 (3-2)
Theory of Automata	3 (3-0)

AI Course Contents

1. Python
2. AI approaches
3. Knowledge representation techniques
4. Rule based programming (OPS-5)
5. Propositional/PredicateLogic
6. Search techniques
7. General Problem Solver (GPS)
8. Text matching (ELIZA)
9. Natural Language processing (STUDENT)
10. Machine Learning basics
11. Prolog

AI Course Learning Objectives (CLOs)

CLO1	Understand key components in the field of artificial intelligence.
CLO2	Analyze artificial intelligence techniques for practical problem solving.
CLO3	Implement classical artificial intelligence techniques.

Reference Books

- Stuart Russell and Peter Norvig, Artificial Intelligence. A Modern Approach, 3rd edition, Prentice Hall, Inc., 2010. 2. Hart, P.E., Stork, D.G. and Duda, R.O., 2001.
- Pattern classification. John Willey & Sons. 3. Luger, G.F. and Stubblefield, W.A., 2009.
- Deep Learning , Bengio Y., Goodfellow I. J., and Courville A, MIT Press, 2016.
- Neural Networks: A comprehensive Foundation, Haykin S., 3rd Edition Pearson Education Press, 2008.

Artificial

‘Artificial’ normally refer to those things which are manmade. Its opposite is ‘natural’ which refers to those things which already exist in this world in their original form without the alterations made by man.

Intelligence

A very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience.

(REF: Mainstream Science on Intelligence reprinted in Gottfredson (1997). Intelligence p. 13)

Examples:

What is the next number in this series: – 1,1,2,3,5,___ (Prediction)

Would you cross a road when a fast car is approaching? (Rational action)

What is Artificial Intelligence?

Artificial intelligence is the science of making machines that can think like humans. It can do things that are considered "smart."

Views/approaches of AI fall into four categories:

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- Top: Thought processes and Reasoning
- Bottom: Behavior
- Left: Human performance
- Right: Ideal concept of intelligence-> **Rationality**

Rationality (A system is rational if it does the “right thing” given what it knows.)

All four approaches to AI have been followed.

Definitions of AI:

Systems that think like humans	Systems that think rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Chamiak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that act like humans	**Systems that act rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) “AI …is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

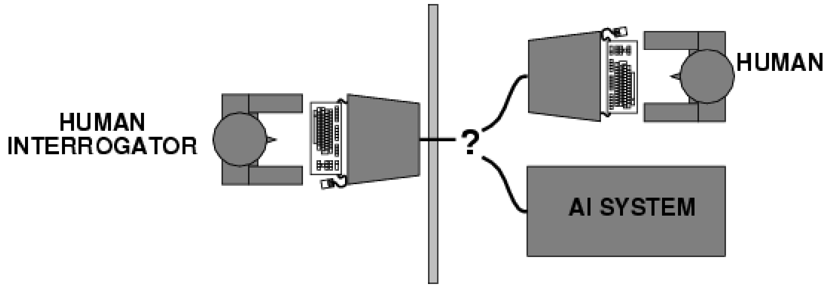
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

1. **Thinking humanly: The cognitive modeling approach**

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. Cognitive Science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of workings of human mind.

2. **Acting humanly: The Turing Test approach**

The Turing Test, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



Suggested following capabilities needed by computer to make it intelligent

i. Natural language processing
   Successful communication in English
ii. Knowledge representation
   To store what it knows or hear
iii. Automated reasoning
   Use the stored information to answer questions and to draw new conclusions
iv. Machine learning
   To adapt to new circumstances and to detect and extrapolate patterns
v. Computer vision
   To perceive objects
vi. Robotics
   To manipulate objects and to move about

Application of the Turing Test is CAPTCHA**:** Completely Automated Public Turing test to tell Computers and Humans Apart.

In 1991 the American philanthropist Hugh Loebner started the annual Loebner Prize competition, promising a $100,000 payout to the first computer to pass the Turing test and awarding $2,000 each year to the best effort. However, no AI program has come close to passing an undiluted Turing test.

3. Thinking rationally: The "laws of thought" approach

The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that syllogism is, irrefutable reasoning processes. His syllogisms provided patterns for argument structures that always yielded correct conclusions when given correct premises—for example, "Socrates is a man; all men are mortal; therefore, Socrates is mortal." These laws of thought were supposed to govern the operation of the mind; their study initiated the field called **logic**. Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them.

4. Acting rationally: The rational agent approach

An agent is just something that acts. Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals. A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Measurement of Human Intelligence:

A test called IQ (Intelligence Quotient) is used to measure the intelligence in humans.

Measurement of Artificial Intelligence:

To check what level of intelligence is present in some machine/software AI measurement levels are used. The broad classes of outcome for an AI test are:

i. Optimal: it is not possible to perform better
ii. Strong super-human: performs better than all humans
iii. Super-human: performs better than most humans
iv. Par-Human: performs parallel to humans
v. Sub-human: performs worse than most humans

Examples

Optimal AI	Strong Super-Human	Super-Human	Par-Human	Sub-Human
Tic-Tac-Toe	Bridge	Backgammon	OCR	OCR
Connect Four	Chess	Crosswords	Image Classification	Handwriting Recog.
Checkers	Go	Driving		Object Recognition
Rubik's Cube	Jigsaw			Translation
	Reversi			Speech Recognition
	Scrabble			NLP
	Quiz Show Question			

In 1945 Turing predicted that computers would one day play very good chess, and just over 50 years later, in 1997, Deep Blue, a chess computer built by the International Business Machines Corporation (IBM), beat the reigning world champion, Garry Kasparov, in a six-game match.